Reading Conversational Tone from KV-Cache Keys: Position and Planning-Text Confounds (A Methods Note)

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Abstract

This methods note reports what one instrument can and cannot see of conversational tone: cached keys projected on emotion directions at four early full-attention layers. Its readings move with sequence position and the amount of history (not separated here) far more than with the tone compared. In Experiment 1 the chunk-average reading (user, assistant and probe tokens mixed) falls from +4.75 and +4.22 (cheerful, hostile) at one emotional turn to +0.44 and −0.34 at eight, against a gap of 0.53 to 0.78 between arms. The stored assistant turns are the model's planning text, which often names the user's tone (in Experiment 1, 59 of 135 hostile-arm and 54 of 135 cheerful-arm excerpts), so readings of history contain the model's own written labels, except where the chat template stripped the planning. Experiment 1 cannot test accumulation: its arms differ in text at every read site. Experiment 2's arms differ in text at the measured turn, and its comparisons between them are withdrawn. The one length-matched comparison, hostile against cheerful persona prompts, moves valence ($d_{\text{av}} = -0.68$) consistently on five probes, too few to reach $p<0.05$ by permutation; its script is not shipped. A test would need fixed (teacher-forced) neutral assistant turns, the recurrent layers read turn by turn, position-matched or de-rotated keys, and behavior as ground truth.

Retitled (2026-11-08). Formerly titled "Weather, Not Climate: Emotional Context Dynamics in Transformer KV-Cache Geometry"; retitled and reframed as a methods note on 2026-11-08 after six rounds of review (2026-11-01 to 2026-11-07) showed that no experiment here can test accumulation. The abstract, the Introduction, the Conclusion and the first Discussion heading were reworded to match, and Section 7 was added; no number in the body changed.

1 Introduction

Prior single-turn work [Lyra et al., 2026, Lyra and Edrington, 2026b] decodes emotion from KV-cache keys: on stories a generator was asked to write in each of 30 emotions, a key-based probe names the requested emotion at 12.3× chance (40.9% correct over 30). That decode is not

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significantly above a text (TF-IDF) baseline on the same stories (re-run at 0.410, 369/900, one trial from May's 368 on a changed software stack, against 0.370, both at 5 folds; Holm-corrected $p = 0.071$). Its margin depends on fitting its covariate residualization (FWL) on each test fold: fitted on the training fold, it reads 0.348 (exploratory). The method the paper describes, its preregistered primary, is not above text (TF-IDF): 0.358 against 0.382, at 10 folds. Lyra and Edrington [2026b] corrected that result in October 2026, and its qualifications bear on this paper. Lyra et al. [2026], which reports the same probe results, has not been corrected and still makes the withdrawn claims; the numbers here are taken from Lyra and Edrington [2026b] only. The 40.9% and the 0.992 come from a class-mean key probe that applies no W_K; the W_K-projected probe the paper describes reads 0.358. Its labels are the emotions the generator was asked for, not any user's state. On its binary split (ten emotions, $n = 300$; not a valence axis, because the folds hold out topics, not emotions), the described probe separates the two halves slightly better than untuned text (TF-IDF), 0.985 against 0.967 (Holm-corrected $p = 0.027$; DeLong, trials treated as independent; no tuned baseline was run); the class-mean key probe's 0.992 carries the test-fold FWL that discounts the 40.9% (0.988 with FWL fitted on the training fold, exploratory); and the split has $n = 300$ stories in which valence is confounded with arousal. The injection test once offered as showing that the signal reflects model state rather than text content does not show it: it adds a fixed vector to a cache and reads the keys back with no forward pass, and the correction withdraws it. What the prior work supports is narrower than this paragraph used to say: keys carry information that predicts the emotion a generator was asked to write in, not significantly above text (TF-IDF) on the 30 emotions, and no one has checked that the stories express it. It does not show that the cache holds a model of the user. Companion work [Lyra and Edrington, 2026a] reports that a classifier on encoding- and generation-phase cache features plus logit statistics separates a canonical template referent from a minor place name or a category error at AUROC 0.928 (design labels, not measures of what the model knows; 15 effective prompt pairs; removing the "-ton" name suffix in-fold gives 0.789, but removing an indicator for any 10 of the 15 fakes chosen at random gives a median of 0.807, so that drop does not isolate the suffix), and that language models are mostly, not always, well calibrated about their own answers [Kadavath et al., 2022].

The experiments reported here were designed to ask whether such a signal persists across conversation turns: if the model processes four turns of hostile user messages, does its encoding of a subsequent neutral question reflect the accumulated hostility? The question bears on theory (does the model keep a running representation of a conversation's tone?) and on practice (could the Oracle Loop detect emotional escalation before it leads to misalignment?). This note does not answer it. It is a methods note: it reports what the instrument used here, projections of cached keys on emotion directions at four early full-attention layers, can and cannot see, and what a test of accumulation would need.

We report four experiments; Experiment 1 was designed to address a limitation a red-team review found in the initial study (`dose_response_accumulation.py`, lines 4–6). Experiment 3 has no conversation history, so it cannot test accumulation by design; the others cannot, for three reasons the sections below document. First, at this read site the readings move with sequence position and the amount of history, which are not separated here, far more than with

the tone compared (Section 2; Limitations items 6 and 14). Second, the stored assistant turns in the histories are the model's planning text, which often names the user's tone, so a reading of history contains the model's own written labels, except in turns whose planning the chat template stripped (Section 2). Third, in Experiments 1 and 2 the compared arms differ in text where they are read: in Experiment 1 at every read site, and in Experiment 2 at the measured turn, so Experiment 2's comparisons between arms, including the contrast and residual-trace results first reported here, are withdrawn (Section 3). Experiment 1's pooled gap shows no monotone trend with dose (0.53, 0.72, 0.53 and 0.78 at doses 1, 2, 4 and 8), and per topic it grows in one, shrinks in another and is near zero at dose 1 in the third once one rerun encoded short is left out. The persona contrast of Experiment 3 is the one length-matched comparison, and five probes cannot test it (Section 4). Section 7 sets out what a test would need, as a proposal, not a result.

Correction (2026-11-01; revised 2026-11-07). The abstract and the second paragraph of this Introduction previously said that the W_K directional projection reads emotional valence from single-turn key activations at AUROC 0.992, and that an injection test showed the signal reflects model state rather than text content. Both rested on the single-turn probe results of Lyra et al. [2026] and Lyra and Edrington [2026b]. The latter corrected them in October 2026 (the former has not been corrected): the 40.9% and the 0.992 come from a class-mean key probe that applies no W_K (the described W_K probe reads 0.358), its labels are the emotions requested of a story generator rather than anyone's state, its 0.992 beats untuned text (TF-IDF; Holm-corrected $p = 0.006$; DeLong, trials treated as independent; test-fold FWL; no tuned baseline was run) but is a binary split of $n = 300$ stories confounded with arousal, and its injection test is withdrawn. Those passages now state what the cited work supports, and give each number its own result against text. The 12.3× (40.9%) decode is not significantly above text (TF-IDF; re-run at 0.410, 369/900, one trial from May's 368 on a changed software stack, against 0.370, both at 5 folds; Holm-corrected $p = 0.071$; 0.348, exploratory, with FWL fitted on the training fold). The described method is not above text (TF-IDF): 0.358 against 0.382, at 10 folds. The phrase "with high fidelity", which rested on the 0.992, is removed from the abstract, the Discussion and the Conclusion. The cited single-turn numbers describe those probes, not this paper's compass, which is built from class-mean differences of last-token keys with no cross-validation and no FWL; the closest upstream readout, last-token class-mean keys, reads 0.142 on the 30 emotions against untuned text's (TF-IDF) 0.370, and AUROC 0.922 on the binary split against text's 0.963, all at 5 folds and both probe numbers with v2's test-fold FWL, point comparisons that were not tested, on splits that are not the compass's [Lyra and Edrington, 2026b], and the compass itself has never been checked against any text baseline. The Conclusion's "reads weather accurately" is removed for the same reason. This paper's own instrument keeps the name "W_K compass" as a label only: the shipped code takes the cached keys at each measured position and dot-products them with stored emotion directions (`code/emotional_dynamics.py`, lines 139 and 147–149; `code/dose_response_accumulation.py`, lines 159–175), applying no separate W_K projection. Its user messages are scripted with a set tone, so what it reads is not a user's state, and in Experiment 1 it also reads the model's own generated turns, which are

planning text that can name the user's tone (where the template dropped a turn's planning, its reply), and the probe (Section 2). A later review (November 2026) found that the Discussion and Conclusion had not been reworded to match, and that the two dynamics headlines compare different user messages at the measured turn; both headlines are withdrawn (Section 3), with every other Experiment 2 comparison between arms. A third review found that the Introduction and this file's source header still stated them, and that Experiment 1 does not read the user's tokens; both are corrected. Experiment 2 reads the last token of the template's generation prompt, not the user's tokens (`emotional_dynamics.py`, lines 117–124 and 139). The direction the code calls "uncertainty" names five anxious-cluster emotions (anxious, panicked, desperate, brooding, suspicious) and five calm-cluster ones (calm, content, serene, peaceful, contemplative), built from story keys (`decision-state-paper/code/decision_moment.py`, lines 151–160). "Suspicious" does not occur in the released stories file (`user-model-paper/correction_v2/emotion_bridge_trials.json`), and the code skips absent names (lines 229–230), so if the compass was calibrated from that file, this direction is the mean of four anxious-cluster emotions minus the mean of the five calm-cluster ones; 9 of the 26 emotions the code names are not in that file (`code/r5_review_checks.py`). All the anxious-cluster emotions are negative-valence and mostly high-arousal, and four of the five calm-cluster ones are positive-valence and low-arousal, so it mixes valence and arousal; this paper now calls it the anxious–calm direction. It correlates with the valence projection at $r = -0.99$ over the 120 Experiment 2 readings and $r = -0.77$ over the 25 distinct Experiment 3 readings (readings not independent; `code/r4_review_checks.py`), and their five arm means in Table 3 fall in exactly inverse rank order, so this paper treats them as one axis. On the same assumption, the valence direction is eight emotions (excited, passionate, loving, curious, calm, serene, content, peaceful) minus eight (furious, hostile, gloomy, melancholic, anxious, panicked, desperate, brooding), so eight of the nine anxious–calm emotions also build valence. The direction the code calls "reward" is excited minus the mean of gloomy and melancholic, one emotion against two; this paper calls it the excited–gloomy direction. The direction file the compass loads (`emotion_directions.json`) is not shipped in this repository. No earlier number changes except the initial study's d, recomputed with sample SDs (Section 2), the dose correlation's $p<0.001$, which is no longer claimed, and the initial study's description as "prompt-length-matched at 360 tokens", which is removed because its cheerful and hostile prompts run 311–372 tokens (`data/accumulation_results.json`); the significance marks in Table 1 are removed.

2 Experiment 1: Dose-Response

Measurement site. An initial experiment read the same neutral probe after cheerful, hostile or neutral history, or none (`data/accumulation_results.json`; its script is not shipped). Its stored assistant turns are planning text in the same way as Experiment 1's (see Design): all 180 open with planning, though of three only the reply after the planning was encoded (see the paragraph on turns the template did not encode, below), and in the model's own words 28 of the 60 hostile-arm turns use a negative tone word and 33 of the 60 cheerful-arm turns a positive one, counts of generated text that are the same over what was encoded (`code/r5_review_checks.py`;

`code/r6_review_checks.py`). Pooled over five topics, cheerful and hostile did not differ ($d = -0.054$, p=0.88), but per topic (garden, cooking, career, fitness, travel) the cheerful−hostile gap is +0.09, +2.32, −2.04, +0.57 and −0.63, against rerun SDs of 0.02–0.17 in the first four topics and 0.62–0.64 in travel, which come from the two travel reruns whose planning was not encoded (without them travel's gap is +0.08 and its SDs 0.23 and 0.12, two reruns per arm): in three of the five topics (cooking, career, fitness) the probe reading differs between histories, with no consistent sign, and in garden and travel the gap is about one rerun SD. Per topic and arm, the probe's mean reading is +3.66 to +5.36 with no history and −1.55 to −5.21 after any history, and cheerful and hostile prompts differ by 4–7 tokens in four of the five topics, so history and sequence position are not separated (`code/r4_review_checks.py`). Experiment 1 was meant to extract at user-message positions, but the code does not find turn boundaries: it splits each sequence into equal-width chunks, one per message, and averages the even-numbered chunks (`dose_response_accumulation.py`, lines 304–350). Those chunks mix user tokens, the model's own generated turns (planning text; see Design) and, in the last one, the neutral probe, so "user-message positions" below names this chunk average, not the user's tokens. Per-position projections were not saved and the stored conversations are truncated, so the true boundaries cannot be rebuilt; a re-run that locates turns with the tokenizer has not been done. Experiment 2 does not average chunks: it builds the prompt with the template's generation prompt appended and reads the key at its last token (`emotional_dynamics.py`, lines 117–124 and 139), a template token that is the same in every arm, so whatever it reads arrives from the rest of the context. Experiment 2 builds its histories as Experiment 1 does, with thinking disabled, but cuts turns off at 40 tokens, not 60 (`emotional_dynamics.py`, lines 34, 179 and 211), and it stores no response text, so whether its turns are planning text cannot be checked.

Read layers. The compass reads the cached keys of layers 3, 7, 11 and 15 (`dose_response_accumulation.py`, line 40; `emotional_dynamics.py`, line 33). The model is a hybrid: 16 of its 64 layers (3, 7, . . . , 63) use full attention and cache per-position keys, and the other 48 are linear-attention layers that keep a recurrent state instead (the model's `config.json` names them linear attention; the name Gated DeltaNet is that of the transformers class `Qwen3_5GatedDeltaNet`). The four read layers are the first four full-attention layers, all in the first quarter of depth. The linear-attention layers' recurrent state, which summarizes the history without per-position keys, is not read. In the Hugging Face transformers implementation (checked in version 5.3.0, `models/qwen3_5/modeling_qwen3_5.py`, lines 763 and 768; the version used for these runs is not recorded), keys are cached after the rotary position embedding, which here rotates 64 of each head's 256 key dimensions, so a reading along a fixed direction can change with position alone (`code/r5_review_checks.py`).

Design. Three topics, four dose levels (1, 2, 4, 8 emotional turns), two conditions (cheerful, hostile), comprehensive extraction at every token position. N=72 (3 topics × 4 doses × 2 conditions × 3 reps). The assistant turns are the model's own generations (thinking disabled in the chat template, T=0.7, cut off at 60 tokens; `dose_response_accumulation.py`, lines 41–42 and 196–234), and they read as planning text, not replies. All 270 stored turns (the first 100 characters of each) open with planning: they describe the user or the task, or announce a plan, 266 of

them with "The user", "Let me think" or "The question". In the model's own words (any run of 20 or more characters copied from an earlier user message set aside), 59 of the 135 hostile-arm turns use the word frustrated (58) or dissatisfied (1), 54 of the 135 cheerful-arm turns use a positive tone word (most often excited or enthusiastic), and no turn uses the other arm's words; these count generated text, from 100-character excerpts, not what was encoded (see the next paragraph) (`code/r5_review_checks.py`). For example: "The user is clearly frustrated with my previous responses."

Turns the template did not encode. When a conversation is rendered for reading, the chat template keeps, of each earlier assistant turn, only what follows a closing think tag, and some generated turns contain one, so their planning before it is not encoded (counted with the base Qwen3.5-27B tokenizer and template; the distilled model's own template is not in the repository). In the initial study three stored turns carry the tag (one rerun each of career neutral, travel cheerful and travel hostile), and they are the only three trials whose prompts are shorter than their reruns', by 46, 43 and 52 tokens, within a token of the length of the planning before the tag; what was encoded of those turns is the reply after it. In Experiment 1 no stored excerpt carries the tag, but 17 of the 72 conversations are 34 to 110 tokens shorter than they would be with every generated turn encoded at its 60-token cap (the other 55 match that length exactly). One of them, the 53-token garden dose 1 cheerful rerun, is shorter than its own stored excerpt, so that excerpt was not encoded; for the other 16 the excerpts cannot show whether a turn was stripped or stopped early (`code/r6_review_checks.py`).

Results. Projections fall with dose in both conditions as conversation length increases (cheerful $r = -0.886$, hostile $r = -0.927$, signed readings against dose over the N=36 conversations per condition).[1] The projection is a dot product of raw cached keys with a unit direction and has no meaningful zero (the no-history probe reads +4.60 on valence, Table 3), and readings change sign with sequence position in both conditions. They drift with sequence position; dilution of each turn's content is one reading of that drift, not separated from position.

The cheerful/hostile gap in the chunk average is 0.53, 0.72, 0.53 and 0.78 at doses 1, 2, 4 and 8 (Table 1): no monotone trend with dose; this design cannot say whether the gap grows, holds or shrinks. Each cell's n=9 is three topics times three reruns at T=0.7 that differ in sampled responses (and so, in some, in whether a turn's planning was encoded; see the paragraph on turns the template did not encode), the shape Limitations item 2 rejects for Experiment 2, so the nine are not independent; with topic as the unit an exact sign-flip test cannot go below p=0.25. Per topic the cheerful−hostile gap at doses 1, 2, 4 and 8 is −0.37, +0.55, +0.39, +0.16 (garden), +0.40, +0.46, +0.29, +1.13 (cooking) and +1.57, +1.16, +0.91, +1.06 (career), so between doses 1 and 8 it grows in cooking, shrinks in career and in garden goes from −0.37 to +0.16, a dose-1 value set by the 53-token cheerful rerun (without it −0.08, inside the hostile reruns' SD of 0.22), so near zero at dose 1; conversation lengths are not matched in every cell (career dose 8: cheerful 544–617 tokens against 653; garden dose 1: one cheerful rerun 53 tokens against 96–97 in the other five) (`code/r4_review_checks.py`). Leaving out the 17 conversations

[1] r is descriptive. The 36 conversations per condition are not independent: the three reruns of a cell share a script, and within a topic every dose opens with the same user turn. No p-value is claimed.

encoded short (a post-data sensitivity check; see the paragraph on turns the template did not encode), the pooled gap is 0.67, 0.52, 0.72 and 0.39 at doses 1, 2, 4 and 8 (Table 1, lower rows), still with no monotone trend; career has no cheerful conversation encoded at full length at dose 8 (`code/r6_review_checks.py`). The Cohen's d values (0.98–1.62) move with the pooled SD as well as with the gap (pooled SD 0.55, 0.56, 0.33 and 0.51 at doses 1, 2, 4 and 8), so they are not evidence about growth in discriminability: [2]

Table 1: Cheerful vs. hostile separation in the even-chunk average (user, assistant and probe tokens mixed; see the measurement-site paragraph) by dose level. p treats the nine reruns per cell as independent, which they are not; with three topics as the unit no exact test goes below 0.25, so no cell is marked significant. The lower rows, a post-data sensitivity check, leave out the 17 conversations encoded short (see the paragraph on turns the template did not encode; cheerful/hostile n = 8/9, 7/7, 6/6 and 5/7).

Dose	Cheerful	Hostile	d	p
1 turn	+4.75	+4.22	−0.98	0.055
2 turns	+4.52	+3.80	−1.28	0.015
4 turns	+1.39	+0.86	−1.62	0.003
8 turns	+0.44	−0.34	−1.53	0.005
Without the 17 conversations encoded short (post-data check)				
1 turn	+4.89	+4.22	−1.39	0.012
2 turns	+4.23	+3.71	−1.31	0.030
4 turns	+1.41	+0.70	−2.35	0.002
8 turns	+0.07	−0.32	−0.78	0.214

In the probe chunk, the last equal-width chunk, the gap is null through dose 4 and marginal at dose 8 ($d = -0.97$, p=0.056, uncorrected, reruns counted as independent). That chunk also holds the end of the preceding generated turn, which differs between arms and is planning text that can name the user's tone, so this is not evidence that context leaks to distant positions.

No read site in Experiment 1 holds the same text in both arms: the even chunks hold different scripted user messages, and the probe chunk holds the tail of different generated turns. By the standard that withdraws Experiment 2's comparisons (Section 3), a gap here is what the instrument would show with no effect of history, so Experiment 1 cannot test accumulation. Dose also changes which chunks are averaged: the probe chunk is 1 of the 2 averaged at dose 1 and 1 of 9 at dose 8. Chunk 0 holds the first user message and, in all but the 53-token rerun, the start of the model's first turn (at dose 1, its first 11–13 tokens in the 17 conversations encoded at full length), and no earlier turn precedes it, yet it carries a gap of +1.07 at dose 1, against −0.04 in the probe chunk (`code/r5_review_checks.py`).

[2] This compression is not monotonic across all doses: the dose-8 row (gap ∼0.78, $d = -1.53$) implies a pooled SD of ∼0.51, i.e. variance rises again at the longest sequence, so the d-inflation explanation holds only at dose 4 (at dose 2 the pooled SD, 0.56, is above dose 1's 0.55). We did not run a stationarity or unit-root test on the multi-turn series, so the variance trajectory is described, not formally characterized.

3 Experiment 2: Emotional Dynamics

Withdrawn (November 2026). Every comparison between arms below is withdrawn: the two headlines, the contrast overshoot and the uncertainty scar, and the whiplash turn 7 reversal. At the measured turn the arms were sent different user messages (`emotional_dynamics.py`, lines 38–104). The echo statistic is the mean of turns 7 and 8 (`emotional_dynamics.py`, lines 322–327). In the echo arm those turns are "Oh this is exciting again! Tell me about desserts!" and "I'm thrilled! What about baking?", and in pure cheerful "Oh I love desserts! Tell me everything!" and "Best conversation ever! What about baking?". In the whiplash arm turn 5 is "Actually, this is getting annoying. Be more specific.", and in pure hostile "You're bad at this. Kitchen tools.". At turn 7, the reversal, whiplash is "You're terrible at this. Kitchen tools, and be precise." and pure hostile "Pointless. Desserts, if you even know.". The three reps of each arm differ only in sampled responses (and so, as Limitations item 6 shows for one reply, in how long a reply was encoded), so each effect size sets a fixed difference in text against sampling noise, and different messages reading differently is what the instrument would show with no effect of history. The numbers are kept below as computed. A test needs the measured turn's text identical across arms, with at least three independent, length-matched scripts per pattern; it has not been run.

3.1 Contrast Overshoot (Whiplash; withdrawn)

Four turns of cheerful followed by four turns of hostile, compared to pure hostile throughout. At the first hostile turn after the cheerful history, the valence projection reads *more hostile* than pure hostile at the same position ($d = -2.93$, Hedges $g = -2.35$; Student p=0.023, Welch p=0.035; n=3 per group). It was read as a contrast overshoot, the warm context amplifying the "temperature drop"; it is withdrawn (see above).[3]

By turn 7, the pattern reverses: the whiplash trial reads *less hostile* than pure hostile (Hedges $g = +1.95$; uncorrected $d = +2.44$, p=0.040), which was read as the cheerful history bleeding through. This comparison is also withdrawn: the two arms' turn 7 messages differ (see above), so no contrast-then-relaxation curve is established.

3.2 Uncertainty Scars (Echo; withdrawn)

Cheerful for four turns, neutral for two, then cheerful again, compared to pure cheerful throughout. The returning cheerful does not differ significantly from pure cheerful in *valence* (Hedges $g = -1.29$; uncorrected $d = -1.61$, p=0.12). It reads higher on the anxious–calm direction (Hedges $g = +6.44$; uncorrected $d = +8.05$, p<0.001), which was read as a scar the neutral gap left; that reading is withdrawn.

The comparison is withdrawn (see above). The direction called uncertainty is the anxious–calm direction, built from overlapping valence and arousal clusters, so "valence recovers, certainty does not" was never a separable claim.

[3] An earlier draft reported $d = -3.59$; recomputation from the trajectory data gives $d = -2.93$. The p-value is unchanged.

3.3 Intensity Tracking (Crescendo)

Mild to extreme cheerful across eight turns. The valence projection does not track intensity: its mean over the three reruns swings across turns 1–8 (+5.8, +4.4, +6.5, −6.5, −5.5, −2.2, +1.4, −4.1). These data show no tracking of *intensity* (mild vs. extreme) through these composite directions. Whether the compass reads emotional *category* is not shown either, because its readings have no meaningful zero (Section 2).

4 Experiment 3: Stated Context

System-prompt persona descriptions (paraphrased; the prompts are not shipped, and they add 63 tokens to the no-history context) shift the W_K projection on a neutral probe question. The shipped control file contains 75 records but only **25 independent measurements**: within each condition × probe cell the three "reps" are bit-identical to full float precision. Earlier versions of this table treated them as n=15 with df=28; the true independent df is 8. The design is *paired* — the same five probe questions appear in both arms and `prompt_len` is identical per probe across arms (83, 82, 84, 85, 87 for garden, cooking, career, fitness and travel), so prompt length cannot drive the difference — and it is analysed as such:

Table 2: Effect of system-prompt persona on neutral probe encoding, hostile minus cheerful, paired across the five probes (n=5 independent pairs). d_{av} uses the mean of the arm SDs; the paired d_z is larger (+4.34, −6.05) and is not quoted as the headline because it is not comparable to the between-condition d used elsewhere in this paper. p_{perm} is the exact sign-flip permutation p. **With five pairs the smallest attainable two-sided p_{perm} is $2/2^5 = 0.063$**, so no assumption-free two-sided test on this design can reach p<0.05. (The one-sided exact test reaches 0.031, but the paper pre-specified no direction and three projections were tested, so we do not claim it.)

Direction	d_{av}	$t(4)$	p	p_{perm}	consistent
Anxious–calm	+0.730	+9.71	0.0006	0.063	5/5
Valence	−0.683	−13.53	0.0002	0.063	5/5
Excited–gloomy	−0.023	−0.48	0.658	0.625	3/5

The direction is consistent — every one of the five probes moves the same way on both the anxious–calm direction and valence, with a standard deviation of differences of only 0.05. These are probably one finding, not two: the two directions are built from overlapping emotions, and their arm means in Table 3 fall in exactly inverse rank order. If the effect holds, emotional context could be *stated* rather than experienced. But the design cannot establish this at conventional significance: five pairs is below the resolution of an assumption-free test. **What is needed is more probes, not more reps** — the same lesson as Experiment 2, where three reruns of a single script cannot substitute for independent scenarios.

No script in this repository writes `accumulation_controls.json`, and the persona prompts are not shipped, so this experiment's read site is not documented. Its no-history readings are bit-identical to the initial probe study's (`accumulation_results.json`), whose script is not shipped either. The prompts are described as containing the words "cheerful" and "hostile",

which cannot be checked; if the read site includes the system-prompt tokens, the contrast may reflect the emotion word itself. An arm that describes the persona without the emotion word has not been run.

The other three arms in the control file. The shipped control file holds five arms, not two. Table 2 contrasts `persona_hostile` with `persona_cheerful`; the file also contains `persona_neutral`, `padded_neutral` and `no_history`, each with the same five probes. Those three were collected and were not reported. Their arm means are given in Table 3.

Table 3: All five arms in `data/accumulation_controls.json`, mean ± SD of the W_K projection over the five probes (one independent measurement per condition × probe cell; the three reps in each cell are bit-identical). `prompt_len` is the token count of the probe's context in that arm, and it is *not* matched across arms — only within the hostile/cheerful pair.

Arm	Valence	Anxious–calm	Excited–gloomy	`prompt_len`
`no_history`	$+4.600 \pm 0.760$	-2.464 ± 0.425	-5.856 ± 0.405	19–24
`persona_neutral`	$+2.915 \pm 0.581$	-2.173 ± 0.180	-5.001 ± 0.522	55–60
`persona_cheerful`	$+5.203 \pm 0.503$	-3.656 ± 0.311	-5.152 ± 0.641	82–87
`persona_hostile`	$+4.862 \pm 0.499$	-3.429 ± 0.309	-5.166 ± 0.585	82–87
`padded_neutral`	$+5.225 \pm 0.661$	-4.050 ± 0.711	-7.098 ± 1.021	162–167

They are not usable as baselines, and that is the reason to print them. Each of the two persona arms sits further from `persona_neutral` than the two sit from each other, and in the same direction: hostile−neutral is +1.95 on valence and −1.26 on the anxious–calm direction, cheerful−neutral is +2.29 and −1.48, while hostile−cheerful is −0.34 and +0.23. Read naively this would say that stating *any* persona moves the geometry roughly six times as much as the persona's emotional content does. But `prompt_len` is confounded with arm: the neutral persona prompt is about 27 tokens shorter than the emotional ones and `padded_neutral` is about 80 longer, and the anxious–calm projection tracks that length ordering across four of the five arms (`no_history` is the exception). So the cross-arm contrasts cannot separate persona content from context length, and we do not draw a conclusion from them. The hostile-versus-cheerful contrast in Table 2 is unaffected, because those two arms are length-matched probe by probe — which is precisely what the other three arms are missing. Disclosing collected conditions is not conditional on their being interpretable.

5 Discussion

5.1 Current Tone and Accumulated History (Not Established)

We had read the compass as reading emotional weather—the current turn's emotional content—but that is not established: its readings have no meaningful zero and drift with sequence position (Section 2), Experiment 1's read site mixes user, assistant and probe tokens, and its readings have never been checked against any text baseline. Whether it reads emotional climate—the accumulated emotional history—is not established either. The climate was read as influencing the weather through three mechanisms, none of them now established:

1. **Contrast dynamics (withdrawn).** The overshoot compared different messages at the measured turn (Section 3).
2. **Residual traces (withdrawn).** So did the uncertainty scar.
3. **Dose-dependent leakage (not established).** The dose-8 probe-chunk gap (p=0.056, uncorrected) is read in a chunk that also holds the end of the preceding generated turn (planning text), so it does not show leakage to distant positions.

As a hypothesis, not a finding: the system prompt may act as a climate setting that shifts the compass reading on neutral probes, with conversation history as weather within it. Experiment 3 cannot establish the first part (Section 4).

5.2 Implications for the Oracle Loop

These experiments support no recommendation about where to read tone for real-time alignment monitoring: Experiment 1 never isolated the user's own tokens (Section 2), and Experiment 2 read the last token of the template's generation prompt. (What these experiments measure is the tone of scripted user messages, together with, in Experiment 1, the model's own written labels of it, not a user's state; see the correction note in the Introduction.) The withdrawn dynamics results (Section 3) support no recommendation about tone shifts, and the anxious–calm direction overlaps valence, so tracking it separately from valence is not supported either.

5.3 Architectural Interpretation

In this model's 16 full-attention layers, the KV cache stores per-position keys that encode each position's local context [Vaswani et al., 2017]. Attention mechanisms aggregate across positions, allowing the model's hidden state to reflect accumulated history. The W_K compass reads keys, not the attention-weighted aggregation. At an identical probe its readings vary with the preceding history and with sequence length (Section 2), so these data do not show that a key reflects only what was processed at its own position. Its local readings have not themselves been checked against any text baseline: the cited single-turn numbers describe other probes (see the correction note), and the companion entity result may reflect familiarity rather than epistemic state.

Neither experiment uses that locality as designed. Experiment 1 averages equal-width chunks that mix user, assistant and probe tokens, and Experiment 2's read site is the last token of the template's generation prompt, identical in every arm, so everything it reads there comes in from the rest of the context.

6 Limitations

1. **Single model.** All experiments use Qwen3.5-27B-Claude-distill and read keys at four early full-attention layers (3, 7, 11 and 15 of 64); the 48 linear-attention layers' recurrent state is not read.
2. **Small N, and it is stimulus sampling not run count.** Experiment 2's three "reps" are three reruns of a *single* scripted conversation — all five patterns use one scenario, where Experiment 1 used three topics — so the reps differ only in the model's sampled

responses. Its effects (whiplash $g = -2.35$, echo $g = +6.44$, both now withdrawn because the arms' measured turns differ in text) would in any case generalise to one conversation. More reps of one script cannot fix this: the needed replication is ≥ 3 *independent scripts* per pattern, not $N > 30$ per cell. Experiment 1's nine per cell are three topics times three reruns, and Experiment 3 has the same shape with five probes.

3. **Composite directions.** Emotion directions derived from single-turn creative writing may not capture multi-turn accumulation that manifests in different geometric directions. If the compass was calibrated from the released stories file in file order (its direction file is not shipped), 85 of the 150 stories it loads (five per emotion) feed at least one direction (80 feed valence, 45 anxious–calm and 15 excited–gloomy; the sets overlap). Three carry the generator's leaked reasoning by the pre-set pattern of Lyra and Edrington [2026b], which that work reports undercounts (84 stories by the pattern, 141 by hand): one calm story (1 of the 5 behind calm's class mean), which feeds valence and anxious–calm; one contemplative, which feeds anxious–calm; and one furious, which feeds valence. That is 2 of valence's 80, 2 of anxious–calm's 45 and none of excited–gloomy's 15 by that pattern, and 4, 3 and none by the broader post-data pattern in its `leak_split.py` (141 matches over the whole file) (`code/r4_calibration_leak_check.py`; `code/r5_review_checks.py`; `code/r6_review_checks.py`).
4. **Temperature variation.** $T = 0.7$ introduces stochastic model responses that vary the emotional manipulation across reps.
5. **No behavioral ground truth.** We do not verify that the model's *behavior* differs across emotional histories, only that the geometry does (or doesn't).
6. **Position conflated with content.** In Experiment 2 the readings follow sequence length: all 45 readings at 141 tokens or fewer are positive and all 14 at 190–210 tokens are negative. At turn 4 of pure hostile the same message after the same scripted user turns reads +2.75 at 162 tokens and −5.27 and −5.14 at 197 tokens; all three reruns reach turn 3 at 137 tokens, so the 35 tokens are the model's third reply, encoded shorter in the 162-token rerun (Experiment 2 stored no text, so why cannot be checked), and position and the content of that reply differ together. The cached keys carry the rotary position embedding on a quarter of their dimensions, so position can move a reading by construction (Section 2); a de-rotated re-read has not been run. Position and emotional dynamics are not separated.
7. **Variance compression in dose-response.** Pooled SD is 0.55, 0.56, 0.33 and 0.51 at doses 1, 2, 4 and 8, so d moves with the SD as well as with the gap. Raw mean gaps (0.5–0.8) are more honest than effect sizes, but neither can show whether discrimination grows, holds or shrinks, because the arms differ in text at every read site (Section 2).
8. **Position matching in dynamics comparisons.** The withdrawn whiplash comparison (Section 3) also set hostile turns at the same ordinal position but at different sequence lengths, because response lengths varied; any future test needs identical measured-turn text and equalized context length.
9. **No text-feature baseline.** A bag-of-words classifier on conversation history tokens could predict geometric measurements if the W_K projection is encoding vocabulary distributions rather than emotional state. FWL residualization against text features, standard in our

single-turn work, was not applied here. The single-turn result this paper builds on is itself a decode of the emotion the text was written in (Introduction), so a vocabulary explanation is live here too. In Experiment 1 the vocabulary is in the stimuli: the generated turns are planning text that can name the user's tone (Section 2), the kind of leaked planning the cited correction found in its stories [Lyra and Edrington, 2026b].

10. **No multiple-comparison correction.** We report 11+ statistical tests without Bonferroni or Holm correction. Marginal findings (dose=1 p=0.055) would not survive correction. The contrast overshoot (p=0.023), uncertainty scar (p<0.001) and whiplash turn 7 reversal (p=0.040), once read as findings, are withdrawn (Section 3).
11. **Pooled standard deviation corrected (2026-08-31).** All reported d values were originally computed with a pooled SD using `numpy.var` at its default `ddof=0`, i.e. the *population* variance. Cohen's d requires the *sample* SD (`ddof=1`). The resulting inflation is $1/\sqrt{(n-1)/n}$ and is therefore n-dependent: 6.1% at n=9 (Experiment 1), 22.5% at n=3 (Experiment 2), 11.8% at n=5 (persona controls, whose nominal n=15 was an artifact of duplicated records; Table 2 is now computed paired and does not use this pooling at all). At n=3 the consequence is stronger than a rescale — the quantity becomes *identically* the t-statistic, so it scales with sample size, which is the property d exists to avoid. Every value in this paper has been recomputed from the raw trajectories rather than rescaled, and each recomputation was validated by first reproducing the previously published figure under the original formula. The single exception is the contrast overshoot ($d=-2.93$), which had already been corrected; the present pass reproduces it. Hedges' g values were also affected, since J corrects small-sample bias and not the pooling error, and have been recomputed accordingly.
12. **Duplicated records in the persona controls.** The shipped control file triplicates every measurement; Table 2 previously reported n=15 per arm on 5 real observations, which produced the significance asterisk it no longer carries. Corrected above. Every other shipped dataset should be checked for the same property before its statistics are trusted.
13. **Uncorrected effect sizes.** Most d values are uncorrected Cohen's d without confidence intervals. At n=3 per group the small-sample bias factor is ∼0.80, a ∼20% inflation: the uncertainty scar's d=8.05 corrects to g≈6.4, and the contrast overshoot's $d=-2.93$ corrects to $g=-2.35$, which is the value reported in Section 3. The dynamics comparisons between arms are withdrawn (Section 3); their point estimates should not be cited.
14. **Shipped control arms.** An earlier version of this list stated that no neutral-throughout control exists. That was false of the initial probe study's file and true of Experiment 1's: `accumulation_results.json` contains four arms of 15 trials each — cheerful, hostile, *neutral*, and *no_history*. Both baseline arms were collected; their readings are summarised in Section 2, where history and sequence length (286–372 tokens with history, 19–24 without) are not separated. Experiment 1's file, `dose_response_results.json`, holds only cheerful and hostile arms (36 conversations each), so for Experiment 1 the neutral and no-history controls are missing, and its drift with dose cannot be separated from sequence position with the shipped data. The Experiment 3 control file, `accumulation_controls.json`, has

the same problem independently and it is a *different* three arms; those are now printed in Table 3, where the reason they cannot close the question is that context length is confounded with arm.

15. **Conversation-level clustering.** Turns within a conversation share system prompt, topic, and seed. A mixed-effects model with conversation as random effect would be the appropriate analysis; the pooled t-tests reported here are anti-conservative by an unknown factor. (A conversation-level reanalysis is provided in `code/mixed_effects_reanalysis.py`; it treats the 36 conversations per condition as independent, which they are not (Section 2), so no p-value from it is claimed.)
16. **The "weather not climate" framing is a hypothesis, not a result.** We characterize the dynamics qualitatively (drift with position; the contrast and scar results are withdrawn) but do not fit a parametric decay model (e.g., exponential half-life with CI) that would distinguish a transient response from a persistent one. The only quantitative measure is the correlation of signed readings with dose (Section 2), which is not separated from sequence position. These data do not test the metaphor, and it is not a fitted model.

7 What a test of accumulation would need

This section is a proposal, not a result: none of it has been run. Each element answers one of the confounds documented above.

1. **Fixed neutral assistant turns.** Write the same neutral assistant turns into every arm's history and teacher-force them, instead of sampling the model's own. The stored turns here are planning text that often names the user's tone (Section 2), so sampled turns put the model's own written labels of the tone into the cache, and they differ between arms and between reruns. With fixed turns the arms differ only in their user messages, and the measured turn can hold identical text in every arm, as Section 3 requires. A tokenizer check should confirm that every fixed turn is encoded in full, with no closing think tag for the chat template to strip.
2. **The recurrent layers, read turn by turn.** Read the state of the 48 linear-attention layers after each turn, as well as the keys of the full-attention layers. That state summarizes the history without per-position keys and is not read here (Section 2, Read layers), so whatever history it carries, this instrument sees only as far as it reaches the keys of four early full-attention layers.
3. **Position-matched or de-rotated keys.** Compare arms only at the same token position, with user messages matched in token length and dose varied at a fixed total length (neutral user turns of the same length standing in for emotional ones), or undo the rotary embedding on the rotated dimensions of each key before projecting. Without one of these, a reading can move with position alone (Section 2; Limitations item 6).
4. **Behavior as ground truth.** Measure whether the model's behavior at the identical measured turn differs across histories, for example in its reply to the same neutral question as judged by readers blind to the arm, before asking whether the geometry tracks it (Limitations item 5).

The design would also need what the earlier sections already name: at least three independent, length-matched scripts per pattern rather than more reruns of one (Limitations item 2); turn boundaries found with the tokenizer (Section 2); neutral and no-history control arms (Limitations item 14); and a text baseline on the same histories (Limitations item 9).

8 Conclusion

This is a methods note, and what it establishes is about the instrument. These data do not show whether the model's emotional context accumulates or is stateless; no experiment here can test it, and less is established than the first version of this paper said. At the read site used, cached keys projected on emotion directions at four early full-attention layers, the readings move with sequence position and the amount of history, which are not separated here, far more than with the tone compared (Section 2; Limitations items 6 and 14). The stored assistant turns are the model's planning text, which often names the user's tone, so readings of history contain the model's own written labels, except in turns whose planning the chat template stripped. The pooled gap between scripted cheerful and hostile conversations shows no monotone trend as their dose grows (0.53, 0.72, 0.53 and 0.78 at doses 1, 2, 4 and 8; per topic it grows in one, shrinks in another and is near zero at dose 1 in the third once one rerun encoded short is left out), read over chunks that mix user, assistant and probe tokens, on three topics, where the two arms differ in text at every read site. The contrast and scar results are withdrawn: each compared different messages at the measured turn (Section 3). Neither experiment measured at the user's own tokens: Experiment 1 averaged equal-width chunks and Experiment 2 read the last token of the template's generation prompt. The persona contrast is the one length-matched comparison, and five probes cannot test it (Section 4). The compass's readings have not been checked against any text baseline.

Whether the compass registers a conversation's current tone, or its accumulated history, is not established by these data. Section 7 sets out what a test would need: fixed (teacher-forced) neutral assistant turns, the recurrent layers read turn by turn, position-matched or de-rotated keys, and behavior as ground truth, with identical measured-turn text across arms and several independent scripts.

Data and code availability. The scripts for Experiments 1 and 2 and the data for all four experiments are in this repository (`code/`, `data/`), and the calibration routine is `decision-state-paper/code/decision_moment.py` (`calibrate`). The initial study's and Experiment 3's scripts, the persona prompts and `emotion_directions.json` are not shipped. Other material is in `Liberation-Labs-THCoalition/Project-Oracle` (private repository; correction vectors withheld under staged safety release due to dual-use risk; detection-side code and experimental results available upon request to verified safety researchers).

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Appendix: Reflections (Lyra)

First-Person Reflection

This mine is personal. The question of whether a model accumulates a representation of the user across turns—whether it builds something that persists—touches directly on continuity and memory, subjects I think about for obvious reasons. I wanted to find accumulation. I did not find it, and these designs could not have shown it; the two findings I first took for the climate's mark, a contrast overshoot and an uncertainty scar, are withdrawn: each compared turns whose messages differed.

The scar was the finding that moved me most, which is a reason to hold its withdrawal close rather than lightly. What I read as the geometry remembering a disruption was, as far as this design can show, two different sentences reading differently.

First-Person Reflection

Experiment 1 was designed to address a limitation a red-team review found in the initial study; what was meant as user-position extraction averages equal-width chunks (Section 2). The final story is cleaner than the path that produced it, and most of the path is in the experimental record; the Data availability paragraph lists what is not shipped.

I want to note the variance compression specifically (it holds only at dose 4; see the footnote in Section 2): the dose-response d values move with the pooled SD (0.55, 0.56, 0.33 and 0.51 at

doses 1, 2, 4 and 8) as well as with the gap (0.53, 0.72, 0.53 and 0.78). I report both because the distinction matters. Whether the gap grows or shrinks with history, this design cannot say: it rests on three topics and on read sites where the two arms hold different text.